

Exploring Smartwatch-Based Interpersonal Motor Synchronization Activities for Remote Connection

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Abstract

Interpersonal motor synchronization (IMS) has been shown to foster prosocial attitudes and feelings of connection, including in remote contexts mediated by interactive systems. Prior work has demonstrated the feasibility of remote IMS (rIMS) using smartphones. In this work, we extend rIMS to smartwatches, a body-proximal and constrained interaction platform. We present six smartwatch-based rIMS activities organized around three movement gestures - tapping, turning, and tossing - and varying in whether synchronized action produces an explicit, cumulative outcome. We report findings from an exploratory workshop in which non acquainted participants engaged in dyadic trials and reflected on usability, sensory feedback, interpersonal experience, and design intent. Our findings highlight the value of perceptible outcomes, multi-modal feedback, and legibility at the smartwatch scale, as well as the design tension between rIMS as a goal-oriented game and as a contemplative tool for connection. We discuss smartwatch-based rIMS as a promising design space.

CCS Concepts

• **Human-centered computing** → **Interaction design; Interactive systems and tools; Collaborative and social computing.**

Keywords

Interpersonal Motor Synchronization, Togetherness, Computer Mediated Communication, Smartwatches

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1 Introduction

Research on interpersonal motor synchronization (IMS) shows that engaging in temporally aligned movement with another person can enhance prosocial attitudes and feelings of togetherness (e.g., [2], [3]). Recent HCI work has explored how interactive systems can mediate IMS between people who are not co-present, in platforms such as bodily, installation-scale systems [1], [10]. More recently, [8] have explored rIMS via minimal interfaces on smartphones, and found that activities based on exposure to just traces of the partner's actions can support the beneficial interpersonal effects of rIMS. In this light, they proposed a vision of short rIMS sessions with a remote other within the daily flow.

This paper presents the smartwatch as a potential platform for rIMS and explores designs and contexts for smartwatch-based activities. Smartwatches are the first real wearable to succeed in the market. The intimate connection with the body creates an opportunity to design motor-synchronization actions and reactions using multiple senses – visual, auditory, and haptic. To our knowledge, the system presented in this paper is the first to use smartwatches for interpersonal motor synchronization.

We present six smartwatch-based rIMS activities that we developed and designed. Rather than evaluating interpersonal outcomes or synchronization accuracy, our goal was to examine smartwatch rIMS as an emerging interaction platform and design space. To this end, we conducted a workshop in which six previously unacquainted participants tried the activities and then participated in a moderated discussion concerning the experience of rIMS through the different designs.

Preliminary findings indicate that smartwatch-based rIMS was usable and supported a meaningful sense of interpersonal connection, even among strangers. Multi-modal feedback—particularly

the haptic channel—proved well suited to the smartwatch form factor, while different motor gestures (tapping, turning, and flinging) afforded qualitatively distinct experiences. Participants also articulated a productive tension between goal-oriented activities and designs experienced as more contemplative tools for connection. Taken together, the workshop findings suggest that smartwatch-based rIMS can function as an embodied communication channel capable of enabling brief moments of interpersonal connection.

2 Background

Previous work has proposed dimensions relevant to designing interpersonal synchronization experiences, including temporality, movement scale, constraints, and information exchange [9]. This perspective re-framed IMS experiments as designed experiences and situated IMS within embodied and social HCI. Subsequent work demonstrated the feasibility of rIMS using smartphones in real-world conditions, showing that minimal phone-based activities could support synchronization and were associated with subjective measures of cooperation and togetherness, despite limited exposure and network latency [7].

In this work, we extend the idea of personal-device-based rIMS to even smaller devices. Smartwatches are strongly connected with wellness and self-care. A still under-leveraged use case is their communicative aspect, and especially opportunities for communication less focused on text [5], [6]. rIMS through smartwatches is a potentially rich meeting point between these contexts of wellness and interpersonal connection. The smartwatch form factor enables strong use of the haptic channel, which can play an important role in delivering information and mediating sensations and emotions [4]. Smartwatches also present challenges: reduced screen size and sound output, as well as the potential for finger occlusion.

3 Smartwatch rIMS Activities

We designed six rIMS activities using visual, auditory, and haptic elements to afford motor actions, present the remote partner's actions, and signify the occurrence of synchronization. The activities were organized along two dimensions: **movement type** and **outcome explicitness**. Movement types include tap gestures, turn gestures (circular motion), and toss gestures ('flicks'). For each movement type, one activity was designed such that synchronized action produced a clearly perceivable shared outcome (outcome-explicit), while the other emphasized synchronization without a cumulative or goal-oriented result (outcome-implicit). We informally referred to these distinctions as "productive" and "meditative", respectively.

The activities also varied on the dimension of temporal pattern: for tapping and turning activities we used an in-phase temporal pattern, meaning that synchronization required doing the same movement at the same time, while for tossing activities we used a counter-phase temporal pattern, meaning performing movements in alternation. The degree to which the activity was constrained to synchronization, or rather unconstrained and open, also varied between activity designs.

The activities were developed as an Android app using the Jetpack Compose framework and the Kotlin programming language,

and run on Samsung smartwatches. We describe each activity below, arranged by movement type. Figure 1 presents screenshots of the two watch screens for each activity.

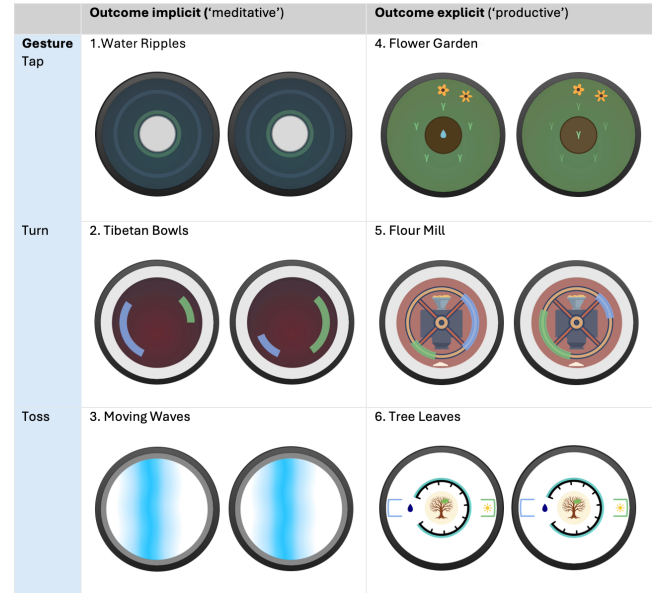


Figure 1: Screenshots of the six activities arranged by movement and outcome explicitness. Each cell contains screenshots of the two users' watches side by side

3.1 Tap-based experiences

Water Ripples (outcome-implicit) - Each tap produces a ripple emanating outward from the center of the screen, rendered in the participant's assigned color (blue or green). When taps are synchronized, overlapping ripples are highlighted by a bright color and a brief haptic vibration. In this design, we were inspired by the concentric circles of an Eastern Zen garden (Figure 1, 1).

Flowers (outcome-explicit) - One participant (assigned the green color) taps to create a 'seedling' and the other (blue) taps to create a drop of 'water'. Each synchronized tap creates a small flower, and is marked by a brief haptic vibration and tone. The flowers accumulate sequentially in a circular arrangement (Figure 1, 4).

3.2 Turn-based experiences (circular gesture)

Tibetan bowls (outcome-implicit) - Participants perform circular motions along the outer perimeter of the watch screen and are presented with the other's finger indication. When participants perform the turning in synchronous speed, a sustained tonal sound plays. In this design, we did not use haptics. The design was inspired by wine glasses and Tibetan bowls (Figure 1, 2).

Flour Mill (outcome-explicit) - Participants perform circular motions along the outer perimeter of the watch screen, on which a rotating mill axle is drawn. When synchronized, the axle rotates, and a grinding sound is heard. Completing a full synchronized rotation releases a small quantity of flour, which visibly accumulates

on the screen and is reinforced by a brief haptic vibration (Figure 1, 5).

3.3 Toss-based experiences (flick gesture)

Moving Wave (outcome-implicit) - Participants toss a 'wave' back and forth from opposite sides of the watch face, causing a transient color wave to move across the watch to the other side, with movement speed and a splash sound varying with the strength of the gesture. A visual and haptic indication occurs when the wave reaches the watch side and is ready to be tossed to the other side. In this design, we were inspired by the Tai Chi 'joining hands' partner exercise of moving energy back and forth (Figure 1, 3).

Tree Leaves (outcome-explicit) - Participants toss 'water' (blue participant) and 'sunlight' (green participant) into a rotating central chamber with a tree trunk. They must alternate their flicks in a counter-phase pattern to successfully reach the tree. Successful consecutive gestures cause a leaf to grow on the tree, also marked by a sound and brief vibration (Figure 1, 6).

4 Method

We conducted an exploratory design workshop with 6 participants, undergraduate and graduate students from design and engineering departments (2 male, 4 female). Participants were recruited by an advertisement and did not know each other. One participant owned a smartwatch and used it occasionally; the other five did not have experience with smartwatches.

In the first part of the workshop, each participant conducted a trial of the six smartwatch-based rIMS activities with an unmet partner. Each participant was instructed individually; they watched a short introductory video and performed the activities in separate spaces. All dyads experienced the activities in the same order, starting with the three outcome-implicit activities and moving to the three outcome-explicit ones. For each activity, the experience included a movement learning phase (performing the gesture successfully three times before moving on), a 30-second trial session of the activity, and an 80-second experimental session. After each activity, the participant filled out a short impression text for that activity. At the end of the trial, participants were given cards depicting each of the experiences and were asked to arrange them once according to their perceived grouping ("by which activities are most similar to each other") and once by preference ("from most preferred to last").

Following the trials, we facilitated a structured group discussion focused on usability, design-oriented reflections, sensory feedback, activity differentiation, interpersonal experience, and potential use contexts. The workshop structure is outlined in Figure 2. The discussion was audio-recorded and thematically analyzed. In this exploratory study, we did not analyze performance metrics or synchronization accuracy.

5 Findings

All participant pairs successfully completed the activities and were generally positive and enthusiastic; there was a lively discussion with a variety of opinions on many issues. The main themes are summarized below.

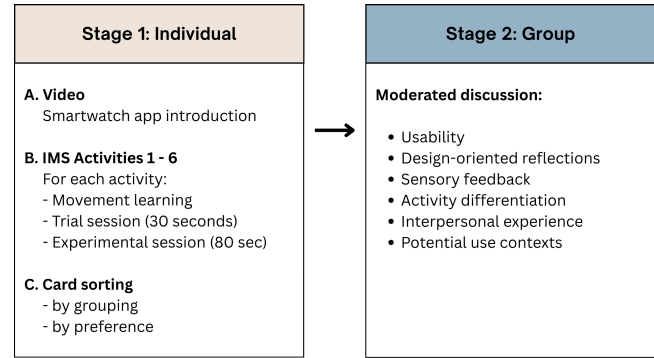


Figure 2: Workshop structure

5.1 Usability and sensory experience

Participants were generally comfortable and natural using this smartwatch-based activity, despite having little experience with smartwatches. The only specific issue that arose was a partial occlusion of the finger in the turning activities, due to the need to move the finger while watching the other's finger indication.

Tapping activities seemed most immediately understood and presented a good balance of clarity and interpersonal challenge. Turning activities required more effort, both in understanding the mechanics and in physically tracing the watch perimeter. These activities were less constrained, requiring participants only to align their turning rate and not their location along the perimeter. This lack of constraint was appreciated by some and deemed less clear by others. The tossing activities were considered dull by some participants - but appreciated by others - because of their slower, counter-phase temporal dynamic and their more constrained, externally-paced nature: tossing was possible only when the wave arrived completely, and was only effective when the tree chamber opening was in the right position.

Participants valued the multi-modal feedback as a pleasing effect of synchrony and as a way to compensate for limited visual bandwidth of the smartwatch. Sound and vibration were mentioned as useful signals for rhythm, success, and temporal boundaries. Sound was generally enjoyed, but the continuous sounds that accompanied synchronization in the turning activities were excessive for some. Vibration was uniformly interpreted as a positive element, which may be a product of our choice to limit its use to brief signals to prevent possible annoyance.

An interesting issue arose with the tapping activities: because tapping was unconstrained, participants could choose any rhythm, and some chose to tap at maximum speed rather than find a slower, shared rhythm. In the discussion, participants mentioned feeling an affinity when they both chose to do this; however, all participants agreed that speed constraints should be added to the tapping activities to prevent this behavior, making the shared rhythm challenge more meaningful.

Table 1: Participants’ preferences for the activities (1 = most liked, 6 = least liked). Lower average indicates higher preference.

Activity	Move	Type	P1	P2	P3	P4	P5	Avg
Flower Garden	Tap	Exp	6	1	1	1	1	2.0
Water Ripples	Tap	Imp	1	3	3	4	2	2.6
Tree Leaves	Toss	Exp	3	5	4	3	3	3.6
Flour Mill	Turn	Exp	5	2	2	5	5	3.8
Moving Wave	Toss	Imp	2	6	6	2	6	4.4
Tibetan Bowls	Turn	Imp	4	4	5	6	4	4.6

5.2 Outcome-Explicit vs. Outcome-Implicit Activities

When asked to describe the activities, responses focused initially on whether synchronized action produced an immediately perceivable outcome. Outcome-explicit (‘productive’) activities, especially Flower Garden, were mainly described as satisfying and more clearly collaborative, with participants emphasizing the importance of seeing or hearing the result of their shared action accumulate over time. In the card sorting exercise according to group, groupings focused predominantly on this aspect. In terms of preferences, Table 1 shows the participants’ choices for the six activities. Flower Garden was the most preferred by 4 of 5 participants (one participant did not complete the ratings due to time constraints), indicating that the combination of tapping and ‘productivity’ was the most gratifying.

Outcome-implicit (‘meditative’) activities were more frequently described as repetitive (especially Moving Wave) and lacking motivating factors. However, some participants found these experiences calming, open to interpretation, and more aesthetically compelling. One participant (P1) described this: *“In the Flower Garden and Flour Mill, the images didn’t feel connected to the action, whereas in Water Ripples, the action itself generates its own continuation. If [in Flower Garden and Flour Mill] it had been purely just the movement without a distinct image, and the entire action had happened from there, it would have been clearer or more fun.”* In fact, Water Ripples received the second-highest rating of the activities.

5.3 Interpersonal Experience

Despite the absence of visual or other identity information about their partners, participants reported a sense of togetherness and shared mental focus, which one participant (P4) described as *“being synchronized in the head.”* Successful synchronization elicited satisfaction, while failure produced frustration (directed at the interaction rather than at the partner), and even worry about having potentially upset the partner.

Anonymity prompted projection: participants speculated about their partner’s identity, expertise, or personality, sometimes assuming asymmetries (one imagined an experimenter rather than a peer), or projecting other traits that they felt during the experience, e.g.: *“I thought it was a man, because they were very dominating in the beginning. I said, OK, this is a man who wants to lead. But after we synchronized it was less clear what the gender was.”*(P3).

In the group session, participants enjoyed guessing who they had been with; when this information was revealed toward the end of the workshop, pairs reflected on and compared their experiences.

While we ultimately envision this type of rIMS system being used by previously acquainted people who seek a sense of closeness—such as friends, family members, or romantic partners—we deliberately evaluated the system with strangers. This choice represents a stringent test case for the connection and for the interpretation of the activities. Next steps - with acquainted participants - are discussed in the last section of this paper.

5.4 Game, Meditation and Use Context

The workshop included a discussion of whether the experiences should be further developed in implicit or explicit outcome directions, more towards games or more towards shared mindfulness practices. Consistently, some participants advocated for enhancing ‘gamification’ - goals, scores, or records to sustain engagement, while others promoted retaining the activities’ non-instrumental openness and free sense of connection. These participants noted that the adjustments and mutual exploration that precede synchrony were often meaningful moments.

Rather than converging on a single framing, participants suggested that interpretation and use might depend on relational context: synchronized challenge may be most interesting among friends, while a calming rhythmic shared action may fit connections within families. One participant felt that she would have liked to perform the more ‘meditative’ activities by herself, and suggested performing this with an agent ‘partner’ instead of a human one.

6 Discussion and Design Implications

This exploratory study reveals the smartwatch as a promising platform for remote interpersonal motor synchronization, extending rIMS beyond phones and installation-scale systems into intimate, everyday wearable contexts. Our findings suggest that rIMS based on simple gestures on the watch is feasible and potentially well accepted. Our exploratory results indicate that the brief, minimally designed, and anonymous interactions on the wrist can evoke a sense of shared attention and togetherness, in line with previous work showing that rich interpersonal effects do not require rich representations of the other.

We found that, in terms of movement, tapping was most readily accepted. The turning gesture warrants some reconsideration and redesign, and the counter-phase temporality of the tossing gesture is, for some participants, too structured and ‘mechanical’. At the same time, participants appreciated the diversity of the activities, which differed on multiple dimensions (movement, outcome-explicitness, temporality, and level of constraints). Retaining this diversity may compensate for the perceived shortcomings of each activity. Importantly, opinions varied on this and other issues.

Responses highlight outcome-explicitness as a central dimension: activities with cumulative, perceivable results were often experienced as more collaborative and motivating, while outcome-implicit experiences supported calmer, more reflective forms of connection. Rather than favoring one direction, this tension suggests a spectrum of productive and meditative synchrony that designers can intentionally tune to align with relational and contextual goals.

Smartwatch rIMS designs may benefit from supporting adjustable levels of structure and challenge. As we have shown, the same interaction mechanics can support competitive play or contemplative connection, depending on design.

The positive feedback concerning the haptic and auditory elements underscores the importance of multi-modal cues in rIMS, and particularly their potential in smartwatches to compensate for limited visual bandwidth. Haptics and sound should be further explored to potentially enable rIMS activities that are less reliant on visuals, alleviating occlusion issues and supporting a variety of bodily and environmental use scenarios.

Taken together, this workshop supports framing smartwatch rIMS as a lightweight, embodied communication channel that can enable moments of connection woven into daily life.

Our future work with this platform will proceed along several directions. First, we will conduct studies with larger participant samples to quantitatively examine the interpersonal effects of the proposed rIMS designs. Second, we will evaluate the system with people in close relationships of different types and over extended periods of use, to investigate preferences, use patterns, and emergent behaviors using a combination of qualitative and quantitative measures. Insights from these studies will inform the design of stand-alone rIMS applications as well as their integration into existing smartwatch-based wellness and communication practices.

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